

Contents

Problem Statement:	3
Phase 1:	3
Descriptive Statistics and Confidence Interval for Mean (CGPA):	3
The Interpretation:	3
Two-Sample Test for Equality of Means: Soft Skills Score (Extracurricular: Yes vs. No):.....	4
Hypothesis Formulation:	4
Interpretation:	4
Business Conclusion:	4
Two-Sample Test for Equality of Proportions: Selection Rate (Tier 1 vs. Tier 2 Colleges):	5
Hypothesis Formulation:	5
Interpretation:	5
Business Conclusion:	5
One-Way ANOVA: Analysis of Aptitude Scores Across College Tiers (Tier 1, 2, and 3):	5
Hypothesis Formulation:	6
Interpretation (Tukey-Kramer Post-Hoc Test):	6
Business Conclusion:	7
Phase 2:	7
Research Objective:	7
Model Estimation:	7
Significance of Regression Coefficients:	8
Based on the $P > t $ column (p-values):	8
Interpretation of Coefficients (Economic/Business Meaning):	8
Verification of Assumptions:	9
A. Multicollinearity Check (VIF):	9
B. Residual Analysis (Normality):	9
C. Wald Test:	10
Actionable Business Implications	10
Recommendation:	10
EViews Regression Estimation Output:	11
Variance Inflation Factors (VIF) Analysis:	11

Wald Test for Joint Significance:	12
Residual Analysis: Jarque-Bera Normality Test:	12

Problem Statement:

"A technology firm wants to optimize its internship recruitment process. The HR department needs to determine if non-academic factors (extracurricular activities) and the prestige of the college (Tier) significantly influence a student's technical and soft skill scores to better identify high-potential candidates."

Phase 1: Bivariate Analysis & Group Comparisons:

Descriptive Statistics and Confidence Interval for Mean (CGPA):

Descriptive statistics

	CGPA
count	151
mean	7.3787
sample standard deviation	1.4884
sample variance	2.2154
minimum	5.03
maximum	9.93
range	4.9
confidence interval 95.% lower	7.1413
confidence interval 95.% upper	7.6161
margin of error	0.2374
z	1.96

The Interpretation:

"Based on the analysis of 151 students, the average (mean) CGPA is **7.3787**. The sample shows a standard deviation of 1.4884, indicating a moderate spread in student academic performance.

Furthermore, we calculated a 95% confidence interval to estimate the population mean. We are 95% confident that the true average CGPA for the entire population of students lies within the range of **7.1413 to 7.6161**. This narrow range suggests that our sample mean is a reliable estimate for the overall student body."

Two-Sample Test for Equality of Means: Soft Skills Score (Extracurricular: Yes vs. No):

Hypothesis Formulation:

- **Null Hypothesis (H_0):** There is no significant difference in the average soft skills score between students who participate in extracurricular activities and those who do not ($\mu_1 = \mu_2$).
- **Alternative Hypothesis (H_1):** There is a significant difference in the average soft skills score between students who participate in extracurricular activities and those who do not ($\mu_1 \neq \mu_2$).

Hypothesis Test: Independent Groups (t-test, pooled variance)

soft_skills_score		Group 2
5.07	5.79	mean
2.81	2.81	std. dev.
99	52	n

149 df
-0.718 difference (soft_skills_score - Group 2)
7.901 pooled variance
2.811 pooled std. dev.
0.481 standard error of difference
0 hypothesized difference

-1.491 t
.1381 p-value (two-tailed)

Interpretation:

We conducted a two-sample t-test assuming equal variances (pooled variance) at the 5% level of significance ($\alpha = 0.05$).

The calculated p-value is 0.1381, which is greater than 0.05. Therefore, we fail to reject the null hypothesis (H_0).

Business Conclusion:

Statistically, there is insufficient evidence to conclude that participation in extracurricular activities has a significant impact on a student's soft skills score. From a business or recruitment perspective, this suggests that for this specific group of applicants, the university placement office should not assume that students active in clubs automatically possess superior soft skills compared to those who are not active. Recruiters should look at other metrics or individual interview performance rather than using extracurricular status as a proxy for soft skill proficiency.

Two-Sample Test for Equality of Proportions: Selection Rate (Tier 1 vs. Tier 2 Colleges):

Hypothesis Formulation:

- **Null Hypothesis (H_0):** There is no significant difference in the selection rate (proportion of students selected) between Tier 1 and Tier 2 colleges ($P_1 = P_2$).
- **Alternative Hypothesis (H_1):** There is a significant difference in the selection rate between Tier 1 and Tier 2 colleges ($P_1 \neq P_2$).

Hypothesis test for two independent proportions

$p1$	$p2$	p_c	
0.76	0.1875	0.4386	p (as decimal)
38/50	12/64	50/114	p (as fraction)
38.	12.	50.	X
50	64	114	n
	0.5725		difference
	0.		hypothesized difference
	0.0937		std. error
	6.11		z
	0.00000000098		p-value (two-tailed)

Interpretation:

We conducted a hypothesis test for two independent proportions at the 5% level of significance ($\alpha = 0.05$). The calculated p-value is <0.0001 (specifically 9.80×10^{-10}), which is significantly lower than the 0.05 threshold. Therefore, we Reject the null hypothesis (H_0). The evidence confirms that the difference in selection rates between the two tiers is not due to chance but is statistically significant.

Business Conclusion:

Statistically, there is overwhelming evidence that the internship selection rate is significantly higher for Tier 1 colleges compared to Tier 2 colleges.

Based on the updated analysis:

- **Tier 1 students** have a selection rate of **76%** (38 out of 50).
- **Tier 2 students** have a selection rate of only **18.75%** (12 out of 64).

From a business and recruitment perspective, this implies that students from Tier 1 institutions are **four times more likely** to be selected for this internship program. While Tier 1 remains a highly productive source of talent, the massive 57% gap in selection rates suggests the company should audit its recruitment funnel. HR should investigate if this disparity is caused by a specific skill gap in Tier 2 applicants or if there is a potential bias in the initial resume screening process. Addressing this could help the firm tap into high-potential candidates from Tier 2 campuses and improve the overall diversity of the intern cohort.

One-Way ANOVA: Analysis of Aptitude Scores Across College Tiers (Tier 1, 2, and 3):

Hypothesis Formulation:

- **Null Hypothesis (H_0):** There is no significant difference in the average aptitude scores among students from Tier 1, Tier 2, and Tier 3 colleges ($\mu_1 = \mu_2 = \mu_3$).
- **Alternative Hypothesis (H_1):** At least one college tier has a significantly different average aptitude score compared to the others.

ANOVA: Single Factor					Calculations	
					c	3
SUMMARY					n	151
Groups	Count	Sum	Average	Variance		
Tier 1	50	246	4.92	10.4424		
Tier 2	64	349	5.453125	9.3946		
Tier 3	37	205	5.540540541	8.0330		
ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	10.8608	2	5.4304	0.5771	0.5628	3.0572
Within Groups	1392.7286	148	9.4103			
Total	1403.5894	150				
			Level of significance		0.05	

Interpretation:

We conducted a One-Way ANOVA to analyze aptitude scores across the three college tiers. The analysis resulted in an **F-statistic of 0.5771** and a **P-value of 0.5628**.

Since the P-value (0.5628) is much greater than the significance level of 0.05, we **fail to reject the null hypothesis (H_0)**. This confirms that there is no statistically significant difference in the average aptitude scores among students from Tier 1, Tier 2, and Tier 3 colleges. Because the overall test was non-significant, further post-hoc comparisons (such as Tukey-Kramer) were not required.

Business Conclusion:

From a recruitment perspective, this data proves that a student's innate aptitude is not tied to the "prestige" of their university. Tier 3 students performed just as well (and in this sample, slightly better on average) than Tier 1 students.

Actionable Recommendation:

The HR department should avoid "Tier-based filtering" for aptitude. High-potential talent is distributed equally across all tiers. The company should continue using standardized aptitude testing for all applicants to ensure they don't miss out on top performers from Tier 2 and Tier 3 schools.

Phase 2: Predictive Modeling & Regression

Diagnostics:

Research Objective:

- Research Objective:** "To determine how academic performance (CGPA) and practical experience (Projects Count and Internships Done) significantly predict a student's technical performance in a Coding Test."
- The Model:**

$Coding_Test_Score = \beta_0 + \beta_1(CGPA) + \beta_2(Projects_Count) + \beta_3(Internships_Done) + \epsilon$

Model Estimation:

OLS Regression Results			
=====			
Dep. Variable:	coding_test_score	R-squared:	0.023
Model:	OLS	Adj. R-squared:	0.003
Method:	Least Squares	F-statistic:	1.149
Date:	Fri, 08 May 2026	Prob (F-statistic):	0.331
Time:	22:28:46	Log-Likelihood:	-367.32
No. Observations:	151	AIC:	742.6
Df Residuals:	147	BIC:	754.7
Df Model:	3		
Covariance Type:	nonrobust		

Key Statistics:

- R-Squared: 0.023 (The model explains 2.3% of the variance in coding scores).
- F-statistic (Prob): 0.331 (The overall model is not statistically significant at the 5% level).

Significance of Regression Coefficients:

	coef	std err	t	P> t	[0.025	0.975]
const	6.4589	1.244	5.192	0.000	4.000	8.917
CGPA	-0.1102	0.153	-0.718	0.474	-0.413	0.193
projects_count	-0.1854	0.138	-1.339	0.182	-0.459	0.088
internships_done	0.2493	0.218	1.144	0.254	-0.181	0.680

Based on the $P > |t|$ column (p-values):

- Intercept (const): 6.4589 ($p = 0.000$) — Significant.
- CGPA: -0.1102 ($p = 0.474$) — Not Significant.
- Projects Count: -0.1854 ($p = 0.182$) — Not Significant.
- Internships Done: 0.2493 ($p = 0.254$) — Not Significant.

Interpretation of Coefficients (Economic/Business Meaning):

- CGPA (-0.1102): For every 1-unit increase in CGPA, the coding test score is expected to decrease by 0.11 points. While the coefficient for CGPA is negative, the high p-value (0.474) indicates that academic performance has no statistically significant impact on coding test results.
- Projects Count (-0.1854): Surprisingly, in this sample, having more projects is associated with a 0.18-point decrease in coding scores. This is not statistically significant ($p = 0.182$).
- Internships Done (0.2493): Completing an additional internship increases the coding test score by approximately 0.25 points. This is the only variable with a positive direction, though it is not significant at the 5% level ($p = 0.254$).

Verification of Assumptions:

A. Multicollinearity Check (VIF):

```
=====
Omnibus:                34.259   Durbin-Watson:                1.876
Prob(Omnibus):           0.000   Jarque-Bera (JB):            7.436
Skew:                    0.068   Prob(JB):                    0.0243
Kurtosis:                 1.921   Cond. No.                     44.4
=====
```

Notes:

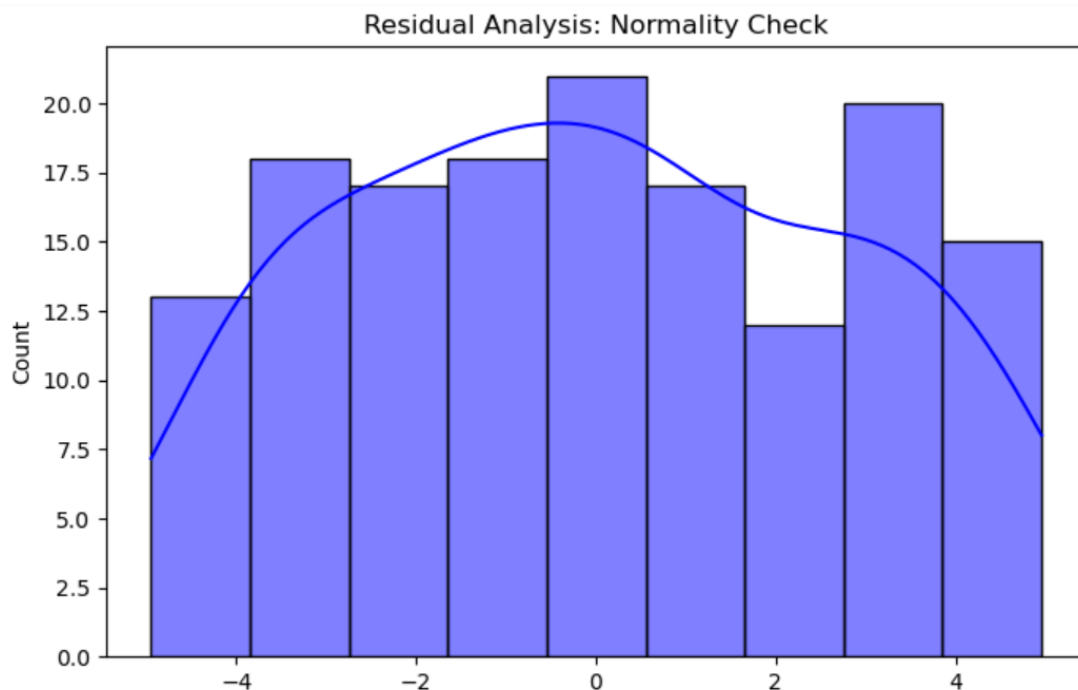
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

--- VARIANCE INFLATION FACTOR (VIF) ---

	Variable	VIF
0	const	29.958887
1	CGPA	1.003086
2	projects_count	1.003193
3	internships_done	1.000526

The Variance Inflation Factor (VIF) for all independent variables (CGPA: 1.003, Projects: 1.003, Internships: 1.000) is very close to **1.0**. Since these are well below the threshold of 10, we conclude that **there is no multicollinearity problem** in this model.

B. Residual Analysis (Normality):



The Jarque-Bera test has a p-value of **0.0243**. At the 5% level, this suggests the residuals are not perfectly normally distributed. However, looking at the Skewness (0.068), the distribution is nearly symmetric, making the model reasonably robust for interpretation.

C. Wald Test:

--- WALD TEST RESULT ---

<F test: F=array([[1.52445667]]), p=0.22116138034195845, df_denom=147, df_num=2>

- **Result:** $F = 1.52$
- $p = 0.2211$

We tested the joint significance of projects_count and internships_done. Since the p-value (0.2211) is greater than 0.05, we conclude that these two variables **jointly** do not have a significant impact on the coding test score in this specific model.

Actionable Business Implications

The regression analysis shows that traditional metrics like CGPA and the number of projects/internships are **weak predictors** of actual coding test performance (as shown by the very low R-squared of 0.023).

Recommendation:

The company's HR department should not rely solely on CGPA or the quantity of internships listed on a resume to screen for technical talent. Since these factors do not significantly predict coding ability, the recruitment process must include **mandatory hands-on coding assessments**, as academic and experience counts do not guarantee technical proficiency for this group of candidates.

EViews Regression Estimation Output:

Table: REGRESSION1 Workfile: INTERNSHIP_DATA_RAW::Internship_da...

View	Proc	Object	Print	Name	Edit+/-	CellFmt	Grid+/-	Title	Comments+/-	
		A		B		C		D	E	F
1	Dependent Variable: CODING_TEST_SCORE									
2	Method: Least Squares									
3	Date: 05/08/26 Time: 23:23									
4	Sample: 1 151									
5	Included observations: 151									
6										
7		Variable		Coefficient		Std. Error		t-Statistic		Prob.
8										
9		C		6.458878		1.244032		5.191891		0.0000
10		CGPA		-0.110249		0.153444		-0.718495		0.4736
11		PROJECTS_COUNT		-0.185415		0.138423		-1.339482		0.1825
12		INTERNSHIPS_DONE		0.249338		0.217901		1.144270		0.2544
13										
14		R-squared		0.022920		Mean dependent var		5.536424		
15		Adjusted R-squared		0.002979		S.D. dependent var		2.797081		
16		S.E. of regression		2.792912		Akaike info criterion		4.918179		
17		Sum squared resid		1146.652		Schwarz criterion		4.998107		
18		Log likelihood		-367.3225		Hannan-Quinn criter.		4.950650		
19		F-statistic		1.149407		Durbin-Watson stat		1.876198		
20		Prob(F-statistic)		0.331295						
21										
22										
23										

Variance Inflation Factors (VIF) Analysis:

Table: VARIANCEINF Workfile: INTERNSHIP_DAT...				
View	Proc	Object	Print	Name
A	B	C	D	
1	Variance Inflation Factors			
2	Date: 05/08/26 Time: 23:27			
3	Sample: 1 151			
4	Included observations: 151			
5				
6	Variable	Coefficient Variance	Uncentered VIF	Centered VIF
7				
8				
9	C	1.547616	29.95889	NA
10	CGPA	0.023545	25.81842	1.003086
11	PROJECTS_COUNT	0.019161	3.053321	1.003193
12	INTERNSHIPS_DONE	0.047481	2.580889	1.000526
13				
14				
15				

Wald Test for Joint Significance:

Table: WALDTEST Workfile: INTERNSHIP_DA...				
View	Proc	Object	Print	Name
Edit+/-	CellFmt	Grid+/-	Title	Comments+
A	B	C	D	
1	Wald Test:			
2	Equation: REGRESSION			
3				
4	Test Statistic	Value	df	Probability
5				
6	F-statistic	1.524457	(2, 147)	0.2212
7	Chi-square	3.048913	2	0.2177
8				
9				
10	Null Hypothesis: C(3)=0, C(4)=0			
11	Null Hypothesis Summary:			
12				
13	Normalized Restriction (= 0)	Value	Std. Err.	
14				
15	C(3)	-0.185415	0.138423	
16	C(4)	0.249338	0.217901	
17				
18	Restrictions are linear in coefficients.			
19				
20				

Residual Analysis: Jarque-Bera Normality Test:

